21CSC529T Inferential Statistics

Unit - 2: PROBABILITY

Syllabus :

Unit - 1: INTRODUCTION TO STATISTICS
Unit - 2: PROBABILITY
Unit - 3: ESTIMATION & HYPOTHESIS TESTING
Unit - 4: HYPOTHESIS TESTING -I
Unit - 5: HYPOTHESIS TESTING -II

Unit 2

- Permutation and Combination
- Types of Probability
- Rules for computing probability
- Marginal probability
- Conditional probability
- Bayes Theorem
- Applications of Bayes Theorem
- Problem-solving in Bayes theorem
- Probability Distributions Binomial & Poisson - Normal Distribution

Permutation and Combination

Permutation

A permutation is an arrangement of objects in a specific order. The sequence or order of the objects is important.

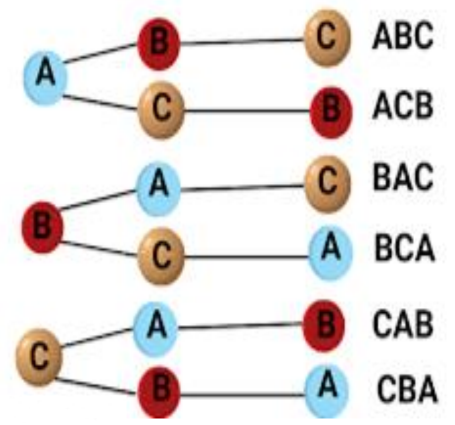
Formula:

$$^{n}P_{r} = \frac{n!}{(n-r)!}$$

n : Total Number of items.

r : Number of items need to select from n numbers

P(n,r): Number of permutations of n objects taken r at a time.



Key Points:

- **Order Matters:** Changing the order of objects creates a different permutation.
- **Examples:** Arranging books on a shelf, creating passwords.
- **Applications in Data Science:** Used in probability calculations, algorithm design, and scenario analysis.

Permutation

A website requires users to create a 4-character password using the digits 0-9. The digits cannot be repeated. How many different passwords can be generated?

- **Solution:**
  - Here, the order of digits matters because "1234" is a different password than "4321."
  - This is a permutation problem where n=10(total digits available) and r=4 (digits in the password).

Answer : Total possible passwords: 5040.

1) A project manager has 5 distinct tasks to assign to 5 time slots in a day. Each task must be completed in one of the slots, and the order of tasks is crucial. How many different ways can the tasks be scheduled?

Solution :

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

This means there are **120 different ways** to arrange the 5 tasks into the 5 available time slots.

2) You have 4 distinct features (F1, F2, F3, F4) that you want to use for training a machine learning model. You are interested in analyzing different ways to order these features and how their order might affect the model's performance. How many different permutations of these features are possible?

Solution:
This is a permutation problem where you are arranging all 4 features in different orders. The number of permutations is calculated as:

$$P(4,4)=4!=24$$

Total Permutations: 24 different ways to order the 4 features.

Combination

A combination refers to a selection of items where the order does not matter. Unlike permutations, changing the order of selected items does not create a new combination.

Formula:

The number of combinations of n items taken r at a time is given by:

$$C(n,r) = \frac{n!}{r!(n-r)!}$$

n: Total number of items.
r: Number of items to choose.

Key Points:

- **Order Doesn’t Matter:** "AB" is the same as "BA".
- **Examples:** Choosing a subset of students from a class, selecting a team members for a game.
- **Applications in Data Science:** Feature selection, experimental design, and data sampling.

Example:

Imagine you have 5 distinct books (A, B, C, D, E) and you want to select 3 of them. The number of possible combinations is calculated as:

Solution :

Total Combinations: 10 ways to choose 3 books from 5.

1) You are organizing a team-building event and need to select 3 team members from a pool of 8 employees. The order in which the team members are selected does not matter.

Solution:

This is a combination problem where you are choosing 3 out of 8 employees, and the order does not matter. You can use the combination formula:

Total Combinations: 56 ways to select 3 team members from 8 employees.

2) You are working on a machine learning project and have 15 distinct features (variables) available in your dataset. You want to select a subset of 5 features to use in your model. The order in which the features are selected does not matter.

Total Combinations: There are **3,003** different ways to select 5 features from the 15 available features for use in your machine learning model.

Probability

Probability

- Probability is a measure for the likelihood of occurrence of an event

$$\text{Probability of an event} = \frac{\text{Number of favourable outcomes}}{\text{Number of all possible outcomes}}$$

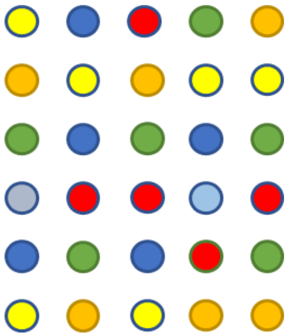
- The probability of any event lies in between 0 to 1
- The sum of all possible events of an experiment is 1
- Probability of an event A is denoted by P(A)

Terminologies in Probability

- **Outcome :** The result of a single trial of an experiment.
- **Event :** It is a set of one or more possible outcomes.
- **Experiment :** An experiment is a process or trial that produces one of several possible outcomes.
- **Sample Space :** set of all possible outcomes of an experiment or collection of event

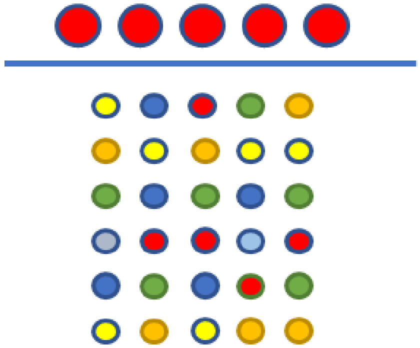
Probability

$$\text{Probability of an event } P(A) = \frac{\text{Number of outcomes in A}}{\text{Number of all possible outcomes}}$$



All possible outcomes

Probability (Ball is red) =



- Probability of complement of an event A is 1 - P(A)
- Probability of an impossible event is 0, i.e. $P(\Phi) = 0$
- For any event B, $0 \leq P(B) \leq 1$
- If event A is a subset of event B ($A \subset B$), then $P(A) \leq P(B)$

Probability

Question:

The new vaccine is to be tested on several patients. There are 5 diabetic patients with same type of diabetes, 9 non-diabetic patients with same heart condition and 11 non-diabetic patients with same liver condition. One patient is randomly chosen. What is the probability that the patient randomly picked is diabetic?

Probability

Solution:

Let A: event that a patient is chosen

Total number of patients = 5 + 9 + 11 = 25

Number of diabetic patients = 5

Probability that a diabetic patients is chosen, i.e. P(A) = $\dfrac{\text{Number of diabetic patients}}{\text{Total number of patients}}$

Probability

Solution:

$$P(A) = \frac{\text{Number of diabetic patients}}{\text{Total number of patients}} = \frac{5}{25} = 0.2$$

What is the probability of choosing non-diabetic patients is , i.e P(A')

The probability of choosing non-diabetic patients is , i.e P(A')

$P(A') = 1 - P(A) = 1- 0.2 = 0.8$

Independence of Events

Independent Events

Two events A and B are **independent** if the occurrence of one event does not affect the probability of the other event occurring.

Mathematical Representation:

$$P(A \cap B) = P(A) \cdot P(B)$$

Key Points:

- Independent events do not influence each other.
- The combined probability is the product of their individual probabilities.

Independent Events

Example: Tossing Two Coins

- **Event A:** Getting Heads on the first coin.
- **Event B:** Getting Tails on the second coin.

  The outcome of the first coin (Heads or Tails) does not influence the outcome of the second coi

  - **Probability:**
    - P(A)=½
    - P(B)=½
    - P(A∩B)=P(Heads on first coin and Tails on second coin) ½ x ½ =¼

Independent event

Question:

Consider rolling a fair die twice :

A: First trial : Occurrence of a number greater than 3

B: Second trail: Occurence of a number lesser than 3

Are the events A and B independent? Why.

Independent event

Solution:

Here the events are:

A: Occurrence of a number greater than 3

B: Occurrence of a number lesser than 3

i.e., $A = \{4, 5, 6\}$ and $B = \{1, 2\}$

$P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$

$P(A \cap B) = P(A).P(B) = \frac{1}{6}$

Thus, events A and B are independent.

A company is running two independent marketing campaigns:

1. **Campaign A**: A digital ad campaign.
2. **Campaign B**: A social media promotion.

The company wants to assess the probability that a customer will make a purchase after being exposed to these campaigns.

- **Probability of making a purchase after seeing Campaign A**: 0.4 (40%).
- **Probability of making a purchase after seeing Campaign B**: 0.3 (30%).

What is the probability that a customer will make a purchase if they are exposed to both Campaign A and Campaign B?

Solution:

Since the two campaigns are independent, the probability of making a purchase after exposure to both campaigns is the product of the individual probabilities:

1. **Probability of making a purchase after seeing Campaign A**: $P(A)=0.4$
2. **Probability of making a purchase after seeing Campaign B**: $P(B)=0.3$
3. **Probability of making a purchase after seeing both Campaigns (A and B)**:
   $P(\text{Purchase} \mid \text{A and B})=P(A)\times P(B)=0.4\times 0.3=0.12$

Therefore, the probability of a customer making a purchase after being exposed to both campaigns is **0.12** or **12%**.

Mutually Exclusive Event

Two events are said to be **mutually exclusive** if they cannot happen at the same time. The occurrence of one event prevents the occurrence of the other.

Mathematical Representation:

$$P(A \cap B) = 0$$
$$P(AUB) = P(A) + P(B)$$

Where A∩B represents both events happening together, which is impossible for mutually exclusive events.

Example: Rolling a Die

- **Event A:** Rolling a 1 (Outcome: {1}).
- **Event B:** Rolling a 2 (Outcome: {2}).

Since you cannot roll both a 1 and a 2 in a single roll of the die, these events are **mutually exclusive**.

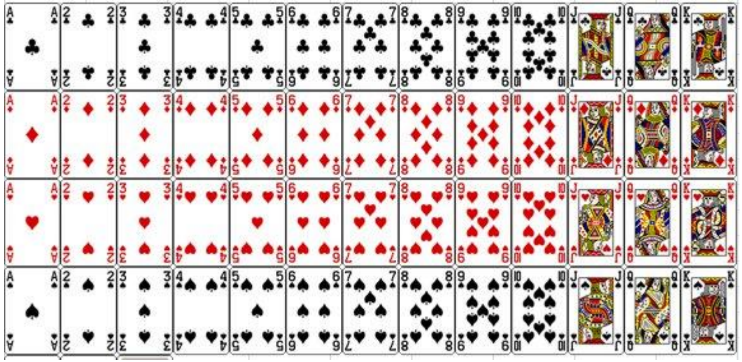
Mutually exclusive

Question:

A card is drawn from a pack of 52 cards. Let events A and B be:

A: Occurrence of a black card

B: Occurrence of a red card



Are the events A and B mutually exclusive? Why.

Solution:

Event A is a black card 26/52 and event B is the occurrence of a red card 26/52.

Implies, A ∩ B is empty. Thus the events A and B mutually exclusive.

A retail company is analyzing the likelihood of customer purchases based on the time of day they shop. They identify two mutually exclusive segments for customers:

1. **Segment A**: Customers who shop in the morning (8 AM - 12 PM).
2. **Segment B**: Customers who shop in the afternoon (12 PM - 4 PM).

It is known from past data that:

- **Probability of a customer shopping in the morning**: 0.45 (45%).
- **Probability of a customer shopping in the afternoon**: 0.40 (40%).

These two segments are mutually exclusive because a customer can only be in one segment at a time; they cannot shop in both time periods on the same visit. What is the probability that a randomly selected customer will shop either in the morning or the afternoon?

Solution:

Since the two segments are mutually exclusive, the probability of a customer shopping in either time period is the sum of the individual probabilities:

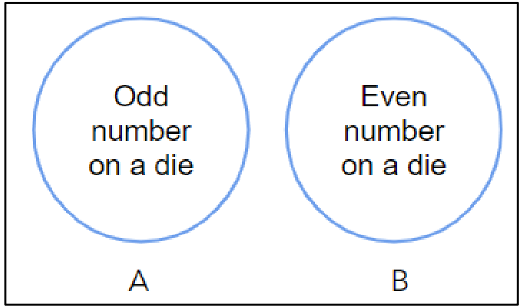
1. **Probability of shopping in the morning**: $P(A)=0.45P(A) = 0.45P(A)=0.45$
2. **Probability of shopping in the afternoon**: $P(B)=0.40P(B) = 0.40P(B)=0.40$
3. **Probability of shopping either in the morning or in the afternoon**:
   $P(\text{Morning or Afternoon})=P(A)+P(B)=0.45+0.40=0.85$

Therefore, the probability that a customer will shop either in the morning or in the afternoon is **0.85** or **85%**.

Independence and mutually exclusive events

- If A and B are independent events with P(A) and P(B) both non-zero, then A and B cannot be mutually exclusive

- If A and B are mutually exclusive with P(A) and P(B) both non-zero, then A and B cannot be independent i.e A and B are dependent



Rules for computing Probability

- If events are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B)$$

- If there is commonality between events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- If the events are Independent

$$P(A \cap B) = P(A) \cdot P(B)$$

- If A and B are not independent, you must use conditional probability:

$$P(A \cap B) = P(A) \times P(B|A)$$

Marginal Probability

- Marginal probability refers to the probability of a single event occurring, regardless of the outcomes of other related events.

- It is obtained by summing (or integrating) the joint probabilities over all possible outcomes of the other variables.

For two events A and B, the marginal probability of A is:

$$P(A) = \sum_{B} P(A \cap B)$$

Example:

- Suppose you have the probability of buying a product (event A) given different income levels (event B). The marginal probability P(A) is the total probability of buying a product, considering all income levels.

Product Bought	High Income	Low Income	Total (Marginal Probability)
Yes	0.3	0.2	0.5
No	0.1	0.4	0.5

Visualization:

- In a contingency table, marginal probabilities are the totals for each row or column.

A company tracks employee performance and job satisfaction levels. Each employee is categorized based on their job satisfaction level and their performance rating. The company has categorized employees into three performance levels (High, Medium, Low) and three satisfaction levels (Satisfied, Neutral, Dissatisfied). Here is the data from a survey of 300 employees:

- **High Performance:** 100 employees
  - **Satisfied:** 70
  - **Neutral:** 20
  - **Dissatisfied:** 10
- **Medium Performance:** 150 employees
  - **Satisfied:** 80
  - **Neutral:** 40
  - **Dissatisfied:** 30
- **Low Performance:** 50 employees
  - **Satisfied:** 10
  - **Neutral:** 15
  - **Dissatisfied:** 25

Questions:

1. **Find the marginal probability that a randomly selected employee has a High Performance rating.**
2. **Find the marginal probability that a randomly selected employee is Satisfied.**

Solution:

1. Marginal Probability of High Performance:

To find the marginal probability of an employee having a High Performance rating, sum the number of employees with High Performance and divide by the total number of employees:

P(High Performance) = Number of High Performance Employees/ Total Number of Employees
= 0.333

2. Marginal Probability of Satisfaction:

To find the marginal probability of an employee being Satisfied, sum the number of employees who are Satisfied across all performance levels and divide by the total number of employees:

P(Satisfied) = Number of Satisfied Employees/ Total Number of Employees
= 0.533

Joint Probability

- **Joint Probability** refers to the probability of two (or more) events occurring simultaneously.
- It is denoted as $P(A \cap B)$, where A and B are two events.
- Represents the likelihood of both events happening together.

Example:

- **Event A**: Drawing a red card from a deck $P(A)=26/52=0.5$
- **Event B**: Drawing a King $P(B)=4/52=0.077$
- **Joint Probability** (Drawing a red card and a King): $P(A \cap B)=0.5\times 0.077=0.0385$

Conditional Probability

- The conditional probability of an event A is the probability that the event will occur given the knowledge that an event B has already occurred

- It is denoted by,

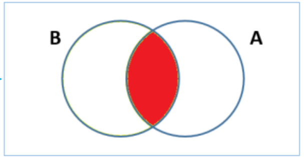
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

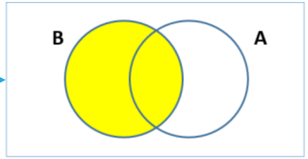
Conditional Probability

- The conditional probability of an event A is the probability that the event will occur given the knowledge that an event B has already occurred

- It is denoted by,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$





Conditional probability

Question:

A pair of fair dice is rolled. If the product of numbers that appear is 6, find the probability that the second die shows an even number?

Conditional probability

Solution:

Let A: the event of getting the product as 6

The ways A can occur: {(1,6), (2,3), (3,2), (6,1)}

Let B: the event that the second die shows an even number

The ways B can occur: {(1,2), (1,4), (1,6), (2,2), (2,4), (2,6), (3,2), (3,4), (3,6), (4,2), (4,4), (4,6), (5,2), (5,4), (5,6), (6,2), (6,4), (6,6)}

First Dice

Second Dice	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Thus, the event that product of the die is 6 and number 2 appears on the dice is $A \cap B$.

$$A \cap B = \{(1,6), (3,2)\}$$

The total number of samples is 36.

Conditional probability

Solution:

The required probability is

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)}$$

$$P(\text{number on the second die is even} \mid \text{the product is 6}) = \frac{\text{P (product is 6 and number on the second die is even)}}{\text{P(the product is 6)}}$$

$$P(\text{number on the second die is even} \mid \text{the product is 6}) = \frac{\frac{2}{36}}{\frac{4}{36}} = \frac{2}{4} = 0.5$$

Let A and B be two events:

Addition theorem: $$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Multiplication theorem:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow P(A \cap B) = P(A \mid B).\,P(B)$$

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

$$\Rightarrow P(A \cap B) = P(B \mid A).\,P(A)$$

Bayes' Theorem

Bayes' theorem

- Conditional probability is the likelihood of an event given that another event has occurred

- Bayes theorem provides a way to update the probability based on the new information

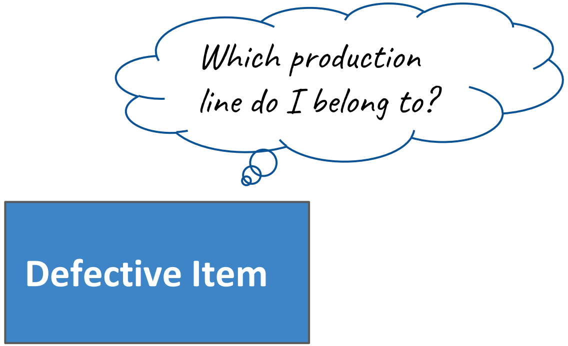
- It is completely based on the conditional probability

- Also known as Bayes' Rule or Bayes' law

Bayes' theorem

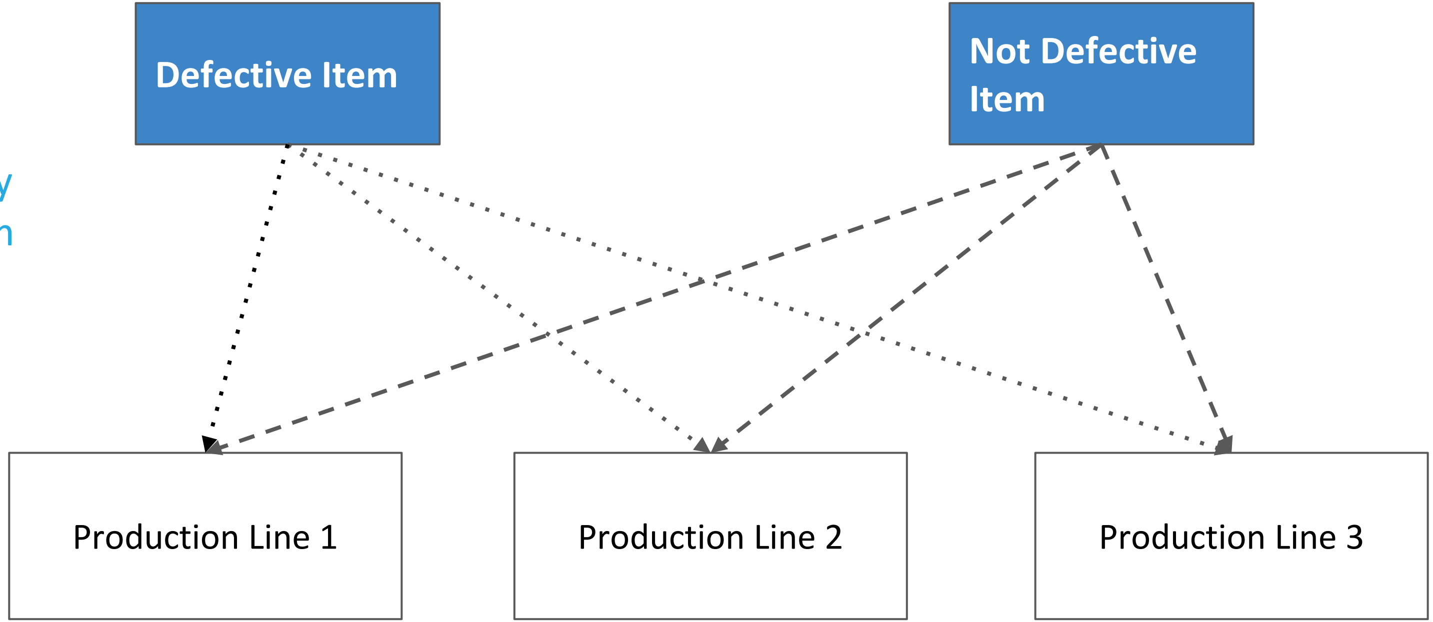
Suppose L_1, L_2 and L_3 are the production lines of an industry and D is the event that the article manufactured is defective. Given the article is defective, we need to find the probability that it was produced on which line.

$$P(L_i \mid D) = \frac{P(L_i).P(D|L_i)}{\sum_{j=1}^{3} P(L_j).P(D|L_j)} \quad \text{for all } i = 1, 2, 3$$



Bayes’ theorem

What is the probability that the defective item chosen was from the production line 1?



$P(L_1) = 0.3$

$P(L_2) = 0.4$

$P(L_3) = 0.3$

Bayes' theorem

Suppose events $A_1, A_2, \ldots, A_n$ form a partition on a sample space Ω of a random experiment and B is any other event such that P(B) > 0 defined on Ω, then

Prior Probabilities

Likelihood

$$P(A_i \mid B) = \frac{P(A_i).P(B|A_i)}{\sum_{j=1}^{n} P(A_j).P(B|A_j)} \quad \text{for all } i = 1, 2, \ldots, n$$

Posterior Probabilities

Evidence

Conditions in Bayes Theorem

1) P(A|B) not equal to P(B|A)
2) There should be dependency between the event.
3) Marginal Probability can’t be zero

Prior probabilities

- Consider three production lines L_1, L_2, L_3 manufacturing the same finished articles
- From the past experience, it is seen that the chance of selecting a production line is 0.3, 0.4, 0.3 respectively
- These probabilities are called the prior probabilities

Posterior probabilities

- Consider three production lines L_1, L_2, L_3 manufacturing the same finished articles

- An article is chosen at random and is found to be defective

- The knowledge of the probability that the defective article is produced by L_1, L_2 or L_3 will help in improving the quality of the product line, i.e. $P(L_1|$ defective article$)$

Likelihood

- Consider three production lines L_1, L_2, L_3 manufacturing the same finished articles
- Likelihood is the conditional probability that the article produced is defective given it is produced on one of the lines L_1, L_2, L_3. i.e. P(defective article | L_1)
- It specifies the chance of producing a defective article of the given production line

Evidence

- Event D is that the article selected is defective, which is not used in the calculation of the prior probability
- In our example, the evidence is the probability of selecting a defective article produced on any of the production lines L_1, L_2 or L_3
- It is also known as marginal probability

Bayes' theorem

Question:

In an armament production station, the explosion can occur due to short circuit, fault in the machinery, negligence of workers. From experience, the chances of these causes are 0.1, 0.3, 0.6 respectively. The chief engineer feels that an explosion can occur with probability:

- 0.3 if there is a short circuit
- 0.2 if there is a fault in the machinery
- 0.25 if the workers are negligent

Given that an explosion has occurred, determine the most likely cause of it?

Bayes' theorem

Solution:

Let the events be defined as

A_1: the explosion can occur due to short circuit

A_2: the explosion can occur due to fault in the machinery

A_3: the explosion can occur due to negligence of workers

The prior probabilities are

$P(A_1) = 0.1$, $P(A_2) = 0.3$, $P(A_3) = 0.6$

Bayes' theorem

Solution:

Let B be the event there is an explosion.

Thus the probabilities are

P(there is an explosion| there is a short circuit) = $P(B \mid A_1) = 0.3$

P(there is an explosion| there is a fault in the machinery) = $P(B \mid A_2) = 0.2$

P(there is an explosion| the workers are negligent) = $P(B \mid A_3) = 0.25$

Bayes' theorem

Solution:

To find: the most likely cause of it given that an explosion has occurred

Thus, using the bayes theorem, to compute the probability that there is a short circuit given an explosion has occurred is computed as

$$P(A_1|B) = \frac{P(A_1).P(B|A_1)}{P(A_1).P(B|A_1) + P(A_2).P(B|A_2) + P(A_3).P(B|A_3)}$$

$$P(A_1|B) = \frac{0.1\times0.3}{0.1\times0.3+0.3\times0.2+0.6\times0.25} = \frac{1}{8} = 0.125$$

Bayes' theorem

Solution:

Similarly,

$$P(A_2|B) = \frac{0.3\times0.2}{0.1\times0.3+0.3\times0.2+0.6\times0.25} = \frac{1}{4} = 0.25$$

$$P(A_3|B) = \frac{0.6\times0.25}{0.1\times0.3+0.3\times0.2+0.6\times0.25} = \frac{5}{8} = 0.625$$

Bayes' theorem

Solution:

Thus, we have

P(there is a short circuit | there is an explosion) = $P(A_1 \mid \mathbf{B}) = 0.125$

P(there is a fault in the machinery| there is an explosion) = $P(A_2 \mid \mathbf{B}) = 0.25$

P(the workers are negligent| there is an explosion) = $P(A_3 \mid \mathbf{B}) = 0.625$

Comparing the probabilities, $P(A_3 \mid \mathbf{B})$ has highest probability. It implies that the negligence of workers is the most likely cause of an explosion in the factory.

Summary

Terminologies	Meaning
Prior Probability	Probability of the event A (P(A))
Posterior Probability	Conditional probability of the event A given event B has occurred (P(A|B))
Likelihood	Conditional probability of the event B given event A has occurred (P(B|A))
Evidence	Probability of the event B (P(B))

Odds

Odds

- Odds of an event is the ratio of the number of observations in favour of an event to the number of observations not in favour of the event

$$\text{odds} = \frac{\text{number of observations in favour of the event}}{\text{number of observations not in favour of the event}}$$

- Probability is sometimes expressed in terms of odds

Odds

Odds in favour of an event A are a:b

Probability of odd in favour of event A

$$P(A) = \frac{a}{a+b}$$

Probability of odd in against of event A

$$P(A') = \frac{b}{a+b}$$

$$P(A') = 1 - P(A)$$

Odds

Question:

Imagine a bag contains **8 red marbles** and **2 blue marbles**. What is Odds in favor of drawing a red marble ?

Solution : The odds in favor of drawing a red marble are the ratio of the number of red marbles to the number of blue marbles.

Odds in favor of red = 4:1

Odds

What is Odds against drawing a red marble:

Solution : The odds against drawing a red marble are the inverse, representing the chance of drawing a blue marble instead of a red one.

Odds against red = 1:4

Imagine a new diagnostic test is developed for detecting a certain disease. In a clinical trial, the test's results show that the patient having the disease. Result of the test to be positive is 7 and negative is 3.

1) What is the odd in favour of patient to be positive.
2) What is the odd in against of patient to be positive
3) What is the probability of patient to be positive

Answer : 1) Odds in favor of a patient with the disease testing positive: **7:3**

2) Odds against a patient with the disease testing positive: **3:7**

3)Probability of a patient with the disease testing positive: **70%**

Summary

Probability of an event is the ratio of number of observations in favour of an event to all possible observations

$$\text{probability} = \frac{\text{number of observations in favour of the event}}{\text{number of observations}}$$

Odds of an event is the ratio of number of observations in favour of an event to number of observations not in favour of the event

$$\text{odds} = \frac{\text{number of observations in favour of the event}}{\text{number of observations not in favour of the event}}$$

Thank you!

Happy Learning :)